



Royal University of Phnom Penh

Faculty of Engineering

Assignment Report

Product Subscription and Billing

Conceptual Model Design

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Contents

1	Overview	3
2	Core Entities	4
2.1	Customer	4
2.1.1	Ownership	4
2.1.2	Billing Control	4
2.1.3	Hierarchical Structures	4
2.1.4	Customer Types	4
2.1.5	Onboarding Compliance	5
2.2	Subscriber	5
2.2.1	Customer Relationship	5
2.2.2	Service Subscription	5
2.2.3	Status Life Cycle	5
2.2.4	Charging and Balances	6
2.2.5	Billing and Usage	6
2.3	Offer	6
2.3.1	Offer Type	6
2.3.2	Charging Model	7
2.3.3	Other Characteristics	7
2.4	Bill Invoice	7
2.5	Payment History	7
2.6	Transactions	8
3	Relational Data Model	8
3.1	Relationship	8
3.2	Database Schema and Constraints	10
4	SQL Queries	11
4.1	Postpaid: Top 10 customers with the highest invoice amounts within the last 3 months	11
4.2	Postpaid: Invoice Amount Distribution by Customer Segment YTD . . .	12
4.3	Postpaid: TOP 10 Customers with Outstanding Balances Over 60 Days .	12
4.4	Postpaid: Monthly Payment Collection Comparison: Current Year vs Last Year	13
4.5	Postpaid: New Postpaid Customers Created (Monthly Trend: Current Year vs Last Year)	14
4.6	Top 10 Recharge Champion over the last 3 months	16
4.7	SMS revenue YTD	16
4.8	Voice revenue YTD	17
4.9	Data Revenue YTD	17
4.10	Revenue Per Bundle - (Monthly Trend: Current Year vs Last Year) . . .	18
5	Sensitive Data and Data Quality Issues	19
6	Appendix: External Files and Links	22

List of Figures

1	High Level Entity Relation of Product Subscription and Billing System .	3
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List of Tables

1	Schema - Purposes	10
2	Constraints Check	10
3	Triggers	11
4	Data Sensitivity	20
5	Data quality Issues	21

1. Overview

This business model outlines the structure of a telecom service provider that manages customers, subscribers, products, balances, billing, and payments. The model supports both prepaid and postpaid subscribers with different charging mechanisms, status transitions, and payment methods.

Below is the high level entity relation:

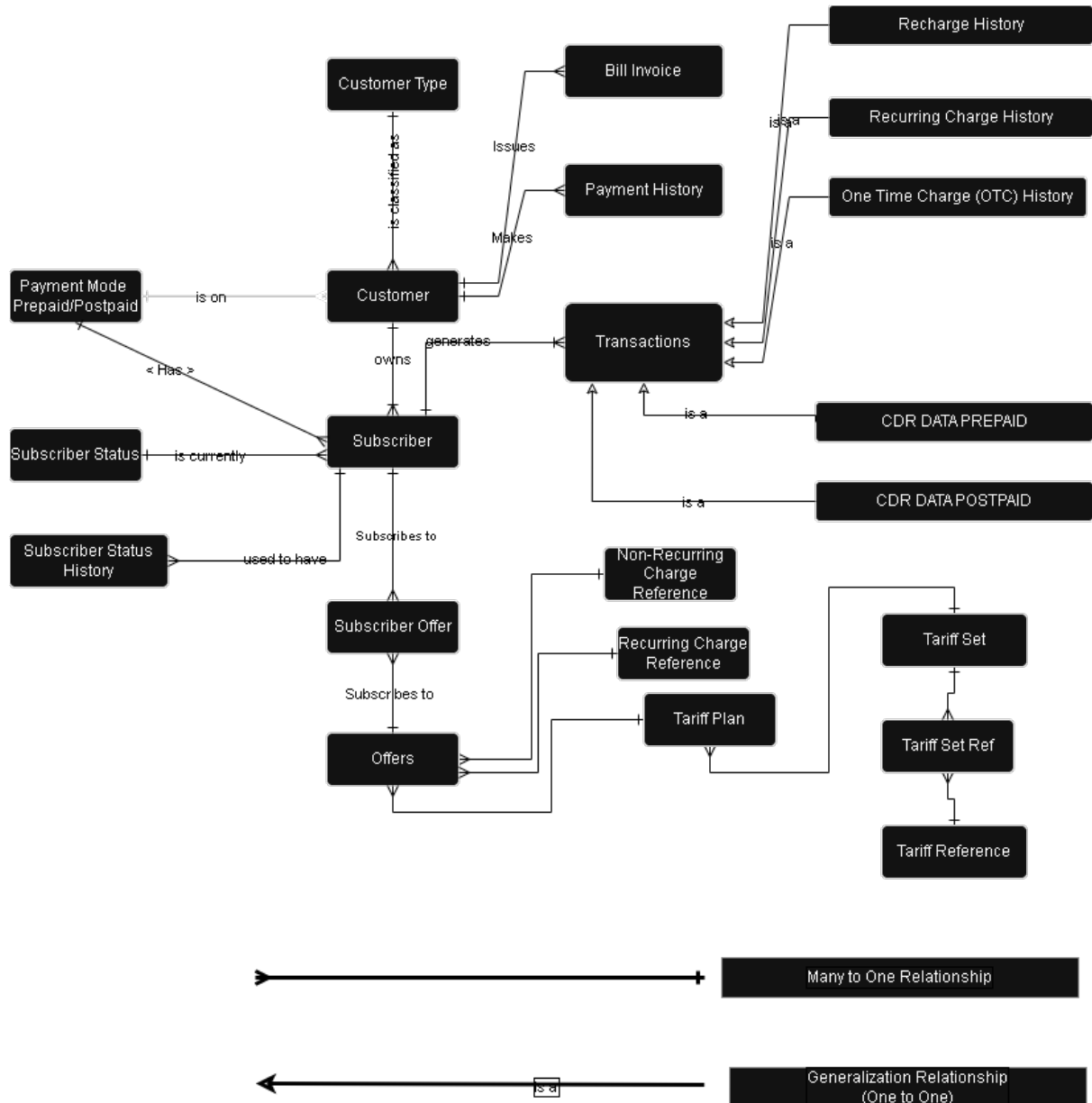


Figure 1: High Level Entity Relation of Product Subscription and Billing System

2. Core Entities

2.1. Customer

The Customer entity represents an account holder in the mobile operator ecosystem. This entity encapsulates both individual and corporate clients and forms the foundation for managing billing, payment, and subscription relationships.

2.1.1. Ownership

- A customer can be either:
 - **Individual (Retail)**: Typically, a single user managing one or more personal mobile services.
 - **Corporate**: Businesses or institutions managing multiple subscriber lines (e.g., employee phones).
- The customer holds the primary authority over billing and payment arrangements for the associated subscribers.

2.1.2. Billing Control

- **Prepaid Model**:
 - Customers fund their subscriber accounts through prepaid balance pool while subscribers could also top-up by themselves.
- **Postpaid Model**:
 - Invoices are issued to the customer on a periodic basis, consolidating all usages from associated subscribers.

2.1.3. Hierarchical Structures

- Applicable mainly to corporate customers.
- Support for hierarchical account management, such as:
 - **Parent Account**: Main billing entity.
 - **Child Accounts**: Departmental or regional sub-accounts under the parent, enabling cost-center tracking and usage segregation.

2.1.4. Customer Types

Classification supports business segmentation and product tailoring:

- **Retail** – Individuals with standard subscriptions.
- **SME** – Small and Medium Enterprises with moderate volume needs.
- **Enterprise** – Large organizations with custom plans and SLAs.

- **Government** – Public sector bodies, such as Prime Minister Office, Ministries. This customer type is always treated prudentially in term of collection and other relation.

2.1.5. Onboarding Compliance

KYC (Know Your Customer) procedures or Credit Assessments are conducted during customer onboarding to validate identity and assess risk (especially for postpaid accounts).

2.2. Subscriber

The Subscriber entity represents an individual service identity associated with a customer, capable of accessing mobile services such as voice, SMS, data, and roaming. Subscribers are the operational unit for service delivery, charging, and usage tracking within the mobile network.

2.2.1. Customer Relationship

- Each subscriber is owned by one customer only.
- A customer may manage multiple subscribers (e.g., family plan, employee lines).

2.2.2. Service Subscription

- A subscriber has to own one Primary offer which defines the type of SIM. For example: Normal Mobile SIM, Machine to Machine SIM (Data Only) etc.
- A subscriber could optionally subscribe other Supplementary offers such as:
 - Data bundles
 - Voice bundles
 - SMS bundles
 - Roaming packages
- These products may be charged through:
 - One-time charges (e.g., 1-day, 7-day data pack)
 - Recurring charges (e.g., weekly, monthly plans)

2.2.3. Status Life Cycle

- A subscriber maintains only one status at a time, selected from a predefined Status entity.
- Typical statuses:
 - Idle – Activated but not yet in use
 - **Active** – Currently in service
 - **Post-active** – Temporarily inactive due to non-payment or expiry

- **Disconnected** – Permanently or administratively removed
- Status transitions depend on:
 - **For prepaid:** Recharge activity and validity extension
 - **For postpaid:** Collection calendar and aging policy

2.2.4. Charging and Balances

- *Prepaid subscribers* maintain rechargeable balances, which fund product subscription and usage.
- Depending on the type of subscription, *postpaid subscriber* could be either consume the balance from the balance pool on account level or monthly pack which is configured to be awarded to each subscriber's balance periodically.
- Multiple balance types can be instantiated per subscriber (e.g., core, bonus, night data, normal data, sms, voice balance) from Primary Offer selection.
- Recharge actions by the subscriber update relevant balance types, mostly Core Balance, which is monetary type.
- Usage records are created against balances, reducing amount accordingly.

2.2.5. Billing and Usage

- For postpaid subscribers, usage is recorded and billed to the owning customer.
- Usage types may include:
 - Pay As You Go: Voice minutes, data volume, SMS count, roaming charges
 - Bundle Charge: Recurring or One-Time-Charge

2.3. Offer

The Offer entity (also referred to as Product) defines a commercial service or package made available for subscription by subscribers.

Offers represent voice, data, SMS, or roaming bundles with pricing, validity, and charging configurations, and form the basis for revenue generation.

2.3.1. Offer Type

There are two types of offer:

- Primary Offer
- Supplementary Offer:
 - Data bundles
 - Voice packs
 - SMS bundles

- Roaming packages
- VAS (Value Added Services such as Ring Back Tone service.)

2.3.2. Charging Model

Offers support multiple charging mechanisms:

- One-time charge (*Non-Recurring Charge Reference*) – Charged once at activation.
- Recurring charge (*Recurring Charge Reference*)– Automatically charged at defined intervals (e.g., daily, weekly, monthly).
- Pay-per-use (*Tariff Plan and its supporting entities*) – Triggered on demand by usage and charged with specific tariff configured on the offer.

2.3.3. Other Characteristics

- Defines Price and Currency to be charged.
- Configurable validity duration (e.g., 1 day, 7 days).
- Eligibility rule - Offer could be restricted or allowed to be subscribed based on:
 - Subscriber type (e.g., Data-only, All-services)
 - Customer type (Retail, SME, etc.)
- Quota and Usage Limits: Define the awards (e.g., 100 minutes, 5Gb of Data)
- Sale Date: Define when to start and to end sale the offer in the system.
- Metadata: OfferID(unique), Offer Name, Description

2.4. Bill Invoice

The Bill Invoice entity represents a formal financial document issued to a postpaid customer for telecom services (Mobile, Broadband..etc.) consumed within a specific billing period. It consolidates all charges from associated subscribers, including product subscriptions, usage, taxes, discounts, and any account-level adjustments.

Each invoice contains details such as:

- Billing period, issue date, and due date
- Itemized charges including recurring fees, overages, and one-time charges
- Payment status (e.g., paid, unpaid, overdue)

2.5. Payment History

The Payment History entity captures all financial transactions made by a customer toward settling their invoices. It provides a historical log of payments including the amount, date, method, and status, supporting financial transparency, reconciliation, and audit compliance.

Each record includes:

- Customer reference and optionally the related invoice (for postpaid)
- Payment amount, date/time, and payment channel (e.g., bank, mobile app, USSD, mobile money)
- Payment method details (e.g., transaction ID for reconciliation with bank or relevant channel)
- Status of the payment transaction (e.g., successful, failed, reversed)

The entity enables tracking of all financial activity across multiple systems and supports reporting on customer payment behavior, collection efficiency, and channel performance.

2.6. Transactions

The Transactions entity records all monetary and service-related operations initiated by a subscriber, including usage events (Voice, Data, SMS), recharges, offer purchases etc.

It serves as a detailed audit trail that supports billing, reconciliation, and customer care inquiries.

This entity supports comprehensive financial logging and real-time updates to subscriber balances, enabling accurate billing, fraud detection, and analytics.

The entity is a bridge to various types of transaction such as:

- Recharge history
- Recurring Charge History
- One Time Charge History
- CDR (Call Detail Record) Prepaid
- CDR Postpaid

3. Relational Data Model

3.1. Relationship

1. Customer & Customer Type

- Relationship: Many-to-One.
- Description: A customer is classified as one customer type, but each customer type can classify many customers.

2. Customer & Payment Mode

- Relationship: Many-to-One.

- Description: A customer is on one payment mode (prepaid/postpaid), but each payment mode can be used by many customers.

3. Customer & Subscriber

- Relationship: One-to-Many
- Description: A customer can own multiple subscribers, but each subscriber belongs to one customer.

4. Customer & Bill Invoice

- Relationship: One-to-Many
- Description: A customer can be issued many bill invoices, but each bill invoice is for one customer.

5. Customer & Payment History

- Relationship: One-to-Many
- Description: A customer can make multiple payments, but each payment history entry is made by one customer.

6. Subscriber & Transactions

- Relationship: One-to-many
- Description: A subscriber can generate many transactions, but each transaction is linked to one subscriber.

7. Subscriber & Subscriber Status

- Relationship: Many-to-One
- Description: One Subscriber could have only one status at a time while each status could be used by many subscribers.

8. Subscriber & Subscriber Status History

- Relationship: One-to-Many
- Description: A subscriber can have multiple past statuses, but each status history belongs to one subscriber.

9. Subscriber & Subscriber Offer Instance

- Relationship: One-to-Many
- Description: A subscriber could have many offer instances but one offer instance could belong only to one subscriber.

10. Subscriber Offer Instance & Offer

- Relationship: Many-to-One
- Description: One Offer ID could be instantiated multiple times but each instance linked only to one offer ID.

11. Transactions & CDR DATA PREPAID (POSPAID)

- Relationship: One-to-One (Specialization)
- Description: a transaction can be a CDR Data Prepaid/Postpaid type, representing an "is-a" relationship (inheritance/subtype).

12. Transactions & Recharge History / Recurring Charge History / One Time Charge (OTC) History

- Relationship: One-to-One (Specialization)
- Description: a transaction can be categorized as one of these specific history types.

3.2. Database Schema and Constraints

To ensure manageability, the database is designed by creating tables in various schema based on each purpose:

Table 1: Schema - Purposes

Schema Name	Purpose
Customer Management (customer_mgmt)	Manage customers and subscribers
Subscription Management (subscription_mgmt)	Track offers and subscriptions
Billing and Transaction (billing_txn)	Payments, bills, charges, and history
Usage Data (usage_cdr)	Usage tracking through CDRs

Across the four database schemas mentioned above, a total of 21 tables have been created, encompassing 132 attributes. Additionally, various constraints and triggers have been implemented to enforce business rules—for example, ensuring that only full payments are accepted for invoices.

Table 2: Constraints Check

Constraints name	Check clause
customer_identity_type_check	((identity_type)::text = ANY ((ARRAY['passport'::character varying, 'national id'::character varying, 'company registration number'::character varying])::text[]))
offer_offer_type_check	(offer_type = ANY (ARRAY[1, 2]))
rc_term_ref_rc_type_check	(rc_type = ANY (ARRAY[1, 2, 3, 4, 5]))
tariff_ref_tariff_type_check	(tariff_type = ANY (ARRAY[1, 2, 3]))

Table 3: Triggers

Table	Trigger name	Descriptions
one_time_charge _history	trg_validate_offer _instance_nrc_term	Validates NRC term for offer instance on insert or update to ensure business rule enforcement.
payment_history	trg_update_invoice_status	Automatically updates invoice status to 'Paid' once full payment is received and recorded.
payment_history	trg_enforce_full_payment	Prevents partial payment by enforcing full amount requirement before transaction is completed.
recurring_charge _history	trg_validate_offer _instance_rc_term	Checks RC term validity for recurring charges during offer instance insert or update.
subscriber_status _history	trg_sync_current_status	Syncs current subscriber status field with latest update for consistency across records.

4. SQL Queries

4.1. Postpaid: Top 10 customers with the highest invoice amounts within the last 3 months

- Data: The query provides the list of top 10 customers who contribute the most of the revenue.
- Importance:
 - focus retention efforts, VIP service, and proactive engagement on these key accounts.
 - with postpaid, higher invoice is one thing while the collection is another thing. Therefore, early warning to tighten credit controls or payment follow-up on big accounts.
 - Big spenders are more likely to be receptive to additional products or services.

```

1      SELECT
2          BI.CUST_ID ,
3          BI.INVOICE_DATE ,
4          BI.TOTAL_AMOUNT
5      FROM
6          BILLING_TXN.BILL_INVOICE BI
7      JOIN
8          CUSTOMER_MGMT.CUSTOMER CUST
9      ON BI.CUST_ID = CUST.CUST_ID
10     WHERE

```

```
11      BI.INVOICE_DATE >= CURRENT_DATE - INTERVAL '3 months'  
12  ORDER BY  
13      BI.TOTAL_AMOUNT DESC;
```

4.2. Postpaid: Invoice Amount Distribution by Customer Segment YTD

- Data: Report provides the progress of invoice amount year to date based on customer segment.
- Importance:
 - Understand which segment drives the most revenue and tailor strategies accordingly.
 - Adjust sales and marketing focus to balance the revenue portfolio.
 - Supports budget forecasting, profitability analysis, and investment decisions.

```
1  SELECT  
2      CUST_TYPE ,  
3      SUM(BI.TOTAL_AMOUNT) BILLED_AMOUNT  
4  FROM  
5      CUSTOMER_MGMT.CUSTOMER CUST ,  
6      billing_txn.BILL_INVOICE BI  
7  WHERE  
8      CUST.CUST_ID = BI.CUST_ID  
9      AND BI.invoice_date >= DATE_TRUNC('year',  
10         CURRENT_DATE)  
11 GROUP BY  
    CUST_TYPE
```

4.3. Postpaid: TOP 10 Customers with Outstanding Balances Over 60 Days

- Data: query lists top 10 customers with total outstanding balances over 60 days
- Importance:
 - focus collection efforts on large overdue amounts to improve cash inflow.
 - senior leadership and auditors need visibility into big aging bill.
 - tailor correct collection strategies such as sending reminder, call or sending remind letter to request for payment.

```
1  SELECT  
2      *  
3  FROM  
4      (  
5          SELECT
```

```
6          CUST_EMAIL ,
7          SUM(BI.TOTAL_AMOUNT) AMOUNT
8      FROM
9          BILLING_TXN.BILL_INVOICE BI,
10         CUSTOMER_MGMT.CUSTOMER CUST
11      WHERE
12          BI.CUST_ID = CUST.CUST_ID
13          and BI.DUE_DATE <= CURRENT_DATE - INTERVAL '
14              60 days'
15          AND UPPER(BI.STATUS) != UPPER('Paid')
16      GROUP BY
17          CUST_EMAIL
18      ) ORDER BY
19          AMOUNT DESC
20      LIMIT
21          10
```

4.4. Postpaid: Monthly Payment Collection Comparison: Current Year vs Last Year

- Data: shows whether monthly collections are improving or declining year-over-year. Grouping by monthly makes collection team stays alert all the time so that action could be taken on-time.
- Importance:
 - trigger deeper analysis or action plans to reverse negative trends.
 - historical trends inform projections for upcoming months.
 - a drop in collections may indicate issues with billing, payment channels, or customer defaults.

```
1 WITH
2     BILL_CUR_YEAR AS (
3         SELECT
4             EXTRACT(
5                 MONTH
6                 FROM
7                     BI.INVOICE_DATE
8             ) BILL_MONTH,
9             SUM(BI.TOTAL_AMOUNT) BILLED_AMOUNT
10        FROM
11            BILLING_TXN.BILL_INVOICE BI
12        WHERE
13            DATE_TRUNC('year', BI.INVOICE_DATE) =
14                DATE_TRUNC('year', CURRENT_DATE)
15        GROUP BY
16            EXTRACT(
17                MONTH
18                FROM
```

```
18                                     BI.INVOICE_DATE
19                                 )
20    ),
21    BILL_LAST_YEAR AS (
22        SELECT
23            EXTRACT(
24                MONTH
25                FROM
26                    BI.INVOICE_DATE
27            ) BILL_MONTH,
28            SUM(BI.TOTAL_AMOUNT) BILLED_AMOUNT
29        FROM
30            BILLING_TXN.BILL_INVOICE BI
31        WHERE
32            DATE_TRUNC('year', BI.INVOICE_DATE) =
33                DATE_TRUNC('year', CURRENT_DATE) -
34                INTERVAL '1 year'
35        GROUP BY
36            EXTRACT(
37                MONTH
38                FROM
39                    BI.INVOICE_DATE
40            )
41    ),
42    BILL_COMPARISON AS (
43        SELECT
44            CUR.BILL_MONTH,
45            CUR.BILLED_AMOUNT BILLED_AMOUNT_CUR_YEAR,
46            LT.BILLED_AMOUNT BILLED_AMOUNT_LAST_YEAR
47        FROM
48            BILL_CUR_YEAR CUR,
49            BILL_LAST_YEAR LT
50        WHERE
51            CUR.BILL_MONTH = LT.BILL_MONTH
52    )
53    select * From bill_comparison
```

4.5. Postpaid: New Postpaid Customers Created (Monthly Trend: Current Year vs Last Year)

- Data: script below generates the comparison list of the payment collection from last year and this year group by month.
- Importance:

—

```
1
2 WITH
3     CUST_CREATED_CURRENT_YEAR AS (
```

```
4          SELECT
5              EXTRACT (
6                  MONTH
7                  FROM
8                      CREATE_DT
9              ) CREATED_MONTH ,
10             COUNT (*) NO_OF_CUSTOMERS
11         FROM
12             CUSTOMER_MGMT.CUSTOMER CUST
13         WHERE
14             DATE_TRUNC('year', CUST.CREATE_DT) =
15                 DATE_TRUNC('year', CURRENT_DATE)
16             and payment_mode = '1'
17         GROUP BY
18             EXTRACT (
19                 MONTH
20                 FROM
21                     CREATE_DT
22             ) ,
23         CUST_CREATED_LAST_YEAR AS (
24             SELECT
25                 EXTRACT (
26                     MONTH
27                     FROM
28                         CREATE_DT
29                 ) CREATED_MONTH ,
30                 COUNT (*) NO_OF_CUSTOMERS
31             FROM
32                 CUSTOMER_MGMT.CUSTOMER CUST
33             WHERE
34                 DATE_TRUNC('year', CUST.CREATE_DT) =
35                     DATE_TRUNC('year', CURRENT_DATE) -
36                     INTERVAL '1 year'
37                 and payment_mode = '1'
38             GROUP BY
39                 EXTRACT (
40                     MONTH
41                     FROM
42                         CREATE_DT
43                 ) ,
44         COMPARISON_LIST AS (
45             SELECT
46                 CUR.CREATED_MONTH ,
47                 CUR.NO_OF_CUSTOMERS CURRENT_YEAR_CUST ,
48                 LT.NO_OF_CUSTOMERS LAST_YEAR_CUST
49             FROM
50                 CUST_CREATED_CURRENT_YEAR as CUR ,
51                 CUST_CREATED_LAST_YEAR as LT
52             WHERE
```

```
52             CUR.CREATED_MONTH = LT.CREATED_MONTH
53
54         )
55     select
56         *
57     FROM
58         COMPARISON_LIST
```

4.6. Top 10 Recharge Champion over the last 3 months

- Data: query shows Top 10 Most Active Rechargers in the Past 3 Months.
- Importance:
 - Helps identify high-value prepaid customers based on recharge behavior.
 - These users likely generate steady and significant revenue, making them ideal for loyalty programs, upselling, or exclusive promotions.

```
1      SELECT
2          CUST.CUST_EMAIL ,
3          SUB.SUBSCRIPTION_NUMBER ,
4          SUM(TRANS.AMOUNT) TOTAL_RECHARGE_AMOUNT
5      FROM
6          BILLING_TXN.RECHARGE_HISTORY RH,
7          BILLING_TXN.TRANSACTIONS TRANS,
8          CUSTOMER_MGMT.SUBSCRIBER SUB,
9          CUSTOMER_MGMT.CUSTOMER CUST
10     WHERE
11         RH.TRANSACTION_ID = TRANS.TRANSACTION_ID
12         AND TRANS.SUBSCR_NO = SUB.SUBSCR_NO
13         AND TRANS.SUBSCR_NO_RESETS = SUB.SUBSCR_NO_RESETS
14         AND SUB.CUST_ID = CUST.CUST_ID
15         AND TRANS.TRANSACTION_DATE >= DATE '2025-07-01' -
16             INTERVAL '3 months'
17     GROUP BY
18         CUST.CUST_EMAIL ,
19         SUB.SUBSCRIPTION_NUMBER
20     order by SUM(TRANS.AMOUNT) desc
21     limit 10
```

4.7. SMS revenue YTD

- Data: report reveals SMS revenue from Jan until current month
- Importance:
 - Enables product owners to monitor SMS revenue KPIs and take appropriate corrective actions in the event of any anomalies.

```
1      SELECT
2          DATE_TRUNC('month', USAGE_START_DT)::date SMS_MONTH,
3          round(SUM(CHARGING_AMOUNT),0) TOTAL_CHARGE_AMOUNT
4      FROM
5          USAGE_CDR.CDR_DATA_PREPAID
6      WHERE
7          SERVICE_TYPE = 'SMS'
8          AND USAGE_START_DT >= DATE_TRUNC('year',
9              CURRENT_DATE)
10         group by DATE_TRUNC('month', USAGE_START_DT);
```

4.8. Voice revenue YTD

- Data: report reveals prepaid voice revenue from Jan until current month
- Importance:
 - Enables product owners to monitor Voice revenue KPIs and take appropriate corrective actions in the event of any anomalies.

```
1      SELECT
2          DATE_TRUNC('month', USAGE_START_DT)::date SMS_MONTH,
3          round(SUM(CHARGING_AMOUNT),0) TOTAL_CHARGE_AMOUNT
4      FROM
5          USAGE_CDR.CDR_DATA_PREPAID
6      WHERE
7          SERVICE_TYPE = 'VOICE'
8          AND USAGE_START_DT >= DATE_TRUNC('year',
9              CURRENT_DATE)
10         group by DATE_TRUNC('month', USAGE_START_DT);
```

4.9. Data Revenue YTD

- Data: report reveals prepaid data revenue from Jan until current month
- Importance:
 - Enables product owners to monitor Data revenue KPIs and take appropriate corrective actions in the event of any anomalies.

```
1      SELECT
2          DATE_TRUNC('month', USAGE_START_DT)::date SMS_MONTH,
3          round(SUM(CHARGING_AMOUNT),0) TOTAL_CHARGE_AMOUNT
4      FROM
5          USAGE_CDR.CDR_DATA_PREPAID
6      WHERE
7          SERVICE_TYPE = 'DATA'
8          AND USAGE_START_DT >= DATE_TRUNC('year',
9              CURRENT_DATE)
10         group by DATE_TRUNC('month', USAGE_START_DT);
```

4.10. Revenue Per Bundle - (Monthly Trend: Current Year vs Last Year)

- Data: below query shows the performance of each bundle by months comparing between current year and last year.
- Importance:
 - Tracks Performance & Trends: Helps identify growth, seasonal patterns, and changes in bundle effectiveness over time by comparing month-to-month revenue performance with the previous year.
 - Measures Revenue Efficiency: Reveals how much revenue is generated per bundle sold, enabling insights into customer value, pricing effectiveness, and potential margin issues.
 - Supports Strategic Decisions: Informs marketing, pricing, and product strategies by highlighting high-performing or underperforming bundles, guiding promotional and product development efforts.

```
1      WITH
2          RC_CHARGE_THIS_YEAR AS (
3              SELECT
4                  O.OFFER_VALUES ,
5                  SUM(AMOUNT) RC_AMOUNT
6              FROM
7                  BILLING_TXN.TRANSACTIONS TRAN,
8                  BILLING_TXN.RECURRING_CHARGE_HISTORY
9                      RCH,
10                 SUBSCRIPTION_MGMT.OFFER O
11              WHERE
12                  TRAN.TRANSACTION_ID = RCH.
13                     TRANSACTION_ID
14                  AND RCH.OFFER_ID = O.OFFER_ID
15                  AND DATE_TRUNC('year', TRAN.
16                     TRANSACTION_DATE) = DATE '
17                     2025-01-01'
18              GROUP BY
19                  O.OFFER_VALUES
20          ),
21          RC_CHARGES_LAST_YEAR AS (
22              SELECT
23                  O.OFFER_VALUES ,
24                  SUM(AMOUNT) RC_AMOUNT
25              FROM
26                  BILLING_TXN.TRANSACTIONS TRAN,
27                  BILLING_TXN.RECURRING_CHARGE_HISTORY
28                      RCH,
29                 SUBSCRIPTION_MGMT.OFFER O
30              WHERE
31                  TRAN.TRANSACTION_ID = RCH.
32                     TRANSACTION_ID
```

```
27         AND RCH.OFFER_ID = O.OFFER_ID
28         AND DATE_TRUNC('year', TRAN.
                TRANSACTION_DATE) = DATE '
                2024-01-01'
29     GROUP BY
30         O.OFFER_VALUES
31     ),
32     LIST_OF_OFFERS AS (
33         SELECT DISTINCT
34             OFFER_VALUES
35         FROM
36             RC_CHARGE_THIS_YEAR
37         UNION
38         SELECT DISTINCT
39             OFFER_VALUES
40         FROM
41             RC_CHARGES_LAST_YEAR
42     ),
43     OFFER_COMPARISON AS (
44         SELECT
45             LO.OFFER_VALUES,
46             THIS.RC_AMOUNT REV_THIS_YEAR,
47             LAS.RC_AMOUNT REV_LAST_YEAR
48         FROM
49             LIST_OF_OFFERS LO
50             LEFT JOIN RC_CHARGE_THIS_YEAR THIS
51                 ON THIS.OFFER_VALUES = LO.
52                 OFFER_VALUES
53             LEFT JOIN RC_CHARGES_LAST_YEAR LAS
54                 ON LAS.OFFER_VALUES = LO.
55                 OFFER_VALUES
56     )
57
58     SELECT
59
60     *
61
62     FROM
63
64     OFFER_COMPARISON;
```

5. Sensitive Data and Data Quality Issues

The billing system is extensive and hosts various types of sensitive data, including customer information, usage records, and financial transactions. Due to its complexity and critical role in revenue assurance, ensuring data integrity and system reliability is paramount.

At the same time, several steps within customer acquisition still rely on manual processes, which inherently introduce the risk of human error. These errors, though often

unintentional, can impact billing accuracy, customer satisfaction, and regulatory compliance.

To address these challenges, the table below outlines the key areas involving manual intervention, along with the corresponding mitigation measures. These actions are designed to maintain operational smoothness, enhance data accuracy, and ensure compliance with regulatory and industry standards.

Table 4: Data Sensitivity

Compliance Factors	Remarks
Customer Proprietary Network Information (CPNI)	The design does have network information such as Call Details Records containing: CellID, Calling, Called Number. However, those used in this design are mocked up.
Personal Health Information (PHI)	This is irrelevant for this system modeling. There is no health information stored in Billing System.
Consumer Financial Information (CFI)	The system contains some of the financial information such as Subscriber's balance and even customer debts. However, all of these sensitive data are allowed to be accessed only by limited personal.
Personally Identifiable Information (PII)	This is totally relevant as per regulatory compliance, KYC (<i>Know Your Customer</i>) has to be fully registered before the subscription is approved to be used for the customer. Data are secured by provided limited access with only approved personal.

Table 5: Data quality Issues

Issues	Description	Mitigation
Customer information	Customer may come multiple times to various branch. Every time he wants to subscribe new service, new entity may be created in the system. This could not ensure uniqueness of one customer in the system	Create a clear new customer acquisition process. Connect to come trust authority such as credit bureau for credit check in case customer insists to have new entity (different customer id) in the system.
Call Details Records(Late)	Late CDR is troublesome. It leads to regulatory compliance issue (In case there'll be requirement from regulator to ensure the CDR availability with specific limited time) as well as billing issue(Late CDR, Late billing which will lead to complaints from the customer).	Proper automated process will be developed to ensure we have the latest CDR in the system.
Customer/ Subscription Information.	Customer information are input by CSR (Customer Service Representative) manually. Therefore, it's always prone to error such as selecting wrong customer type, selecting unwanted payment mode, tax number, wrong products..etc.	With Postpaid service, normally, we do the proforma invoices so that relationship manager could contact their new customers to verify the information before the first bill.

In summary, this conceptual model design takes into account several key compliance and data sensitivity considerations, particularly with respect to Personally Identifiable Information (PII), Customer Proprietary Network Information (CPNI), and Consumer Financial Information (CFI). While Personal Health Information (PHI) is not applicable in this context, the model ensures regulatory alignment, especially for KYC requirements and limited access to sensitive data.

Beyond compliance, data quality poses operational challenges such as customer duplication, late Call Detail Records (CDRs), and manual input errors by customer service representatives. To mitigate these, structured processes have been proposed — including a clear new customer acquisition flow integrated with credit authority validation, automation of CDR ingestion to meet regulatory timeliness, and pre-bill verification via proforma invoices to catch subscription data entry issues. These combined efforts reinforce data governance, enhance system reliability, and maintain smooth day-to-day operations while ensuring full compliance with regulatory standards.

6. Appendix: External Files and Links

- **Product Subscription and ERD Details**

File Name: ProductSubscription_And_Billing_ERD_Details.png [Link]

- **Presentation Deck**

File Name: Product_Subscription_Billing_Conceptual_Modeling_03072025.pptx
[Link]

- **Report**

File Name: ProductSubscription_And_Billing_Conceptual_Design.pdf [Link]

- **Full Database Backup**

File Name: Full_Database_Backup_07072025.sql [Link]